

Statistics: Chi-Square Test for Independence



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DIRECTIONS

Use the chi-square formulas: $E = (\text{row total} \times \text{col total}) / \text{grand total}$; $\chi^2 = \sum (O-E)^2/E$; $df = (\text{rows}-1)(\text{cols}-1)$. If $\chi^2 > \text{critical value}$, reject H_0 and conclude the variables are not independent.

- 1 Find the degrees of freedom for a 2 $\times$ 3 contingency table:

$$df = (r - 1)(c - 1)$$

Answer: _____

- 2 Find the expected frequency E:

$$E = \frac{\text{row total} \times \text{col total}}{\text{grand total}} = \frac{40 \times 30}{100}$$

Answer: _____

- 3 Compute the χ^2 contribution for one cell:

$$O = 15, \quad E = 12$$

Answer: _____

- 4 Find the degrees of freedom for a 3 $\times$ 4 contingency table:

$$df = (r - 1)(c - 1)$$

Answer: _____

- 5 Find the expected frequency E:

$$E = \frac{60 \times 50}{200}$$

Answer: _____

- 6 Compute total χ^2 for these four cells:

$$\frac{(12 - 10)^2}{10} + \frac{(8 - 10)^2}{10} + \frac{(18 - 20)^2}{20} + \frac{(22 - 20)^2}{20}$$

Answer: _____

- 7 At $\alpha = 0.05$, $df = 2$, critical value = 5.991. Make a decision:

$$\chi^2 = 7.32 > 5.991$$

Answer: _____

- 8 Compute the χ^2 contribution for one cell:

$$O = 25, \quad E = 20$$

Answer: _____



Answer Key & Solutions

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TEACHER NOTES

$E = (\text{row total} \times \text{col total}) / \text{grand total}$. $\chi^2 = \text{sum of } (O-E)^2/E \text{ over all cells}$. $df = (r-1)(c-1)$. Reject H_0 if $\chi^2 > \chi^2_{\text{critical}}$ (from table at α and df). All E values should be ≥ 5 for the test to be valid.

- 1 Find the degrees of freedom for a 2 times 3 contingency table:

$$df = (r - 1)(c - 1)$$

$$= df = (2 - 1)(3 - 1) = 2$$

$$df = (\text{rows}-1)(\text{cols}-1) = (1)(2) = 2.$$

- 2 Find the expected frequency E :

$$E = \frac{\text{row total} \times \text{col total}}{\text{grand total}} = \frac{40 \times 30}{100}$$

$$= E = 12$$

$$E = (\text{row total} \times \text{col total}) / \text{grand total} = 1200/100 = 12.$$

- 3 Compute the χ^2 contribution for one cell:

$$O = 15, \quad E = 12$$

$$= \frac{(15 - 12)^2}{12} = \frac{9}{12} = 0.75$$

$$\text{Each cell contributes } (O-E)^2/E. \text{ Here } (3)^2/12 = 9/12 = 0.75.$$

- 4 Find the degrees of freedom for a 3 times 4 contingency table:

$$df = (r - 1)(c - 1)$$

$$= df = (3 - 1)(4 - 1) = 6$$

$$df = (2)(3) = 6. \text{ More cells} \rightarrow \text{more degrees of freedom.}$$

- 5 Find the expected frequency E :

$$E = \frac{60 \times 50}{200}$$

$$= E = \frac{3000}{200} = 15$$

$$E = (\text{row total} \times \text{col total}) / \text{grand total} = 3000/200 = 15.$$

- 6 Compute total χ^2 for these four cells:

$$\frac{(12 - 10)^2}{10} + \frac{(8 - 10)^2}{10} + \frac{(18 - 20)^2}{20} + \frac{(22 - 20)^2}{20}$$

$$= 0.40 + 0.40 + 0.20 + 0.20 = 1.20$$

$$\text{Sum all cell contributions: } 4/10 + 4/10 + 4/20 + 4/20 = 0.4+0.4+0.2+0.2 = 1.20.$$



- 7 At $\alpha = 0.05$, $df = 2$, critical value = 5.991.
Make a decision:

$$\chi^2 = 7.32 > 5.991$$

= Reject H_0 — not independent

Since $\chi^2 = 7.32$ exceeds the critical value 5.991, reject H_0 at $\alpha = 0.05$.

- 8 Compute the χ^2 contribution for one cell:

$$O = 25, \quad E = 20$$

$$= \frac{(25 - 20)^2}{20} = \frac{25}{20} = 1.25$$

$$(O - E)^2 / E = (5)^2 / 20 = 25 / 20 = 1.25.$$

